Geometric Algebra and Maxwell's Equations: A Comprehensive Mathematical Reference

Geometric algebra unifies the inner and outer products of vectors into a single invertible geometric product, enabling all four of Maxwell's equations to be written as the single equation $\nabla F = J$. This formulation, first developed by David Hestenes in 1966 and refined by the Cambridge group in the 1990s, reveals the deep geometric structure of electromagnetism while simplifying computation. The framework builds on 19th-century work by Grassmann, Hamilton, and Clifford, and provides a coordinate-free language that naturally encompasses scalars, vectors, bivectors, rotors, and spinors within one algebraic system. (Springer)

This reference covers the algebraic foundations of geometric algebra (GA), constructs the algebras G(3) and G(1,3), derives Maxwell's equations in both formulations, and develops advanced applications including Lorentz transformations, the stress-energy tensor, and spinor theory.

Part I: Foundations of geometric algebra

1.1 Historical lineage from Grassmann to Hestenes

Geometric algebra synthesizes three major 19th-century mathematical innovations. **Hermann Grassmann** published *Die lineale Ausdehnungslehre* in 1844, introducing the exterior (wedge) product $a \wedge b = -b \wedge a$ as a coordinate-free algebraic method for computing with subspaces. **William Rowan Hamilton** discovered quaternions on October 16, 1843, creating a four-dimensional algebra with basis $\{1, i, j, k\}$ satisfying $i^2 = j^2 = k^2 = ijk = -1$ that elegantly encodes three-dimensional rotations.

William Kingdon Clifford unified these two frameworks in his 1878 paper "Applications of Grassmann's Extensive Algebra," combining Grassmann's exterior algebra with Hamilton's notion of directed numbers that square to scalars. (arxiv) (Wikipedia) Clifford named the result "geometric algebra," (Wikipedia) but died at age 33 in 1879 before fully developing it. The system lay largely dormant for nearly a century. (Wikipedia) **David Hestenes** revived geometric algebra for physics (Wikipedia) with his 1966 monograph *Space-Time Algebra*, reformulating the Dirac equation and Maxwell's equations in the Clifford algebra Cl(1,3). (Wikipedia) His 1984 book with Garret Sobczyk, *Clifford Algebra to Geometric Calculus*, extended the framework to include differentiation and integration. (Springer) (Wikipedia) Chris Doran, Anthony Lasenby, and Stephen Gull at Cambridge further developed applications from the 1990s onward, (DNB) culminating in *Geometric Algebra for Physicists* (2003). (Wikipedia)

1.2 The geometric product and its decomposition

The central object in geometric algebra is the **geometric product** of vectors, defined axiomatically for vectors a, b in a real vector space V equipped with a symmetric bilinear form: (Wikipedia)

- **Associativity**: $a(bc) = (ab)c$

- **Distributivity**: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$
- **Contraction**: $a^2 = aa = a \cdot a \in \mathbb{R}$ (the quadratic form)

From these axioms alone, two fundamental identities follow. Adding and subtracting ab and ba: unicamp

$$a \cdot b = \tfrac{1}{2}(ab + ba) \qquad \text{(inner product — symmetric, scalar-valued)}$$

$$a \wedge b = \tfrac{1}{2}(ab - ba) \qquad \text{(outer product — antisymmetric, bivector-valued)}$$

The geometric product therefore decomposes as: unicamp

$$\boxed{ab = a \cdot b + a \wedge b}$$

This single equation is the cornerstone of geometric algebra. The inner product measures collinearity (magnitude of parallel components), while the outer product measures the oriented area swept out by the two vectors. arxiv arxiv For **parallel vectors** ($a \parallel b$), the wedge product vanishes and $ab = a \cdot b = ba$ (vectors commute). For **orthogonal vectors** ($a \perp b$), the dot product vanishes and $ab = a \wedge b = -ba$ (vectors anticommute). arxiv Wooster

The geometric product is **invertible** for any nonzero vector: if $a^2 \neq 0$, then $a^{-1} = a/a^2$. arxiv +2 This invertibility is a key advantage over the cross product and wedge product alone, neither of which is invertible.

For two vectors with angle θ between them, the geometric product takes the polar form:

$$ab = |a||b|(\cos\theta + \hat{B}\sin\theta) = |a||b|\, e^{\hat{B}\theta}$$

where $\hat{B}$ is the unit bivector of the plane containing a and b. This Euler-like formula connects the geometric product directly to rotations.

1.3 Blades, multivectors, and grade structure

The outer product generalizes to higher grades. A **k-blade** is the outer product of k linearly independent vectors:

$$B_k = a_1 \wedge a_2 \wedge \cdots \wedge a_k$$

representing an oriented k-dimensional subspace. A **multivector** is a general element of the algebra, expressible as a sum of blades of different grades. PlanetMath +3 In an n-dimensional space, the grade structure is: Study.com

Grade k	Name	Geometric meaning	Dimension C(n,k)
0	Scalar	Magnitude	1
1	Vector	Directed line segment	n
2	Bivector	Oriented plane element	n(n−1)/2
3	Trivector	Oriented volume element	n(n−1)(n−2)/6
⋮	⋮	⋮	⋮
n	Pseudoscalar	Oriented n-volume	1

The total dimension of the algebra is $\mathbf{2^n}$ (sum of all binomial coefficients). Every multivector M decomposes uniquely into homogeneous components:

$$M = \langle M \rangle_0 + \langle M \rangle_1 + \langle M \rangle_2 + \cdots + \langle M \rangle_n$$

where $\langle M \rangle_k$ denotes the **grade-k projection**. For the geometric product of a grade-r blade A_r with a grade-s blade B_s:

$$A_r B_s = \langle A_r B_s \rangle_{|r-s|} + \langle A_r B_s \rangle_{|r-s|+2} + \cdots + \langle A_r B_s \rangle_{r+s}$$

The lowest-grade part is the generalized inner product, and the highest-grade part is the outer product:

$$A_r \cdot B_s = \langle A_r B_s \rangle_{|r-s|}, \qquad A_r \wedge B_s = \langle A_r B_s \rangle_{r+s}$$

1.4 The algebra G(3): basis elements and multiplication table

The geometric algebra of three-dimensional Euclidean space, **G(3) = Cl(3,0)**, is generated by three orthonormal basis vectors $\{e_1, e_2, e_3\}$ satisfying:

$$e_i e_j + e_j e_i = 2\delta_{ij}$$

This gives $e_i^2 = 1$ and $e_i e_j = -e_j e_i$ for $i \neq j$. The algebra is **8-dimensional** Instituto de Matemática with the basis: unicamp

Grade	Basis elements	Count
0	1	1
1	e_1, e_2, e_3	3
2	e_{12}, e_{23}, e_{31}	3
3	$I = e_{123}$	1

where $e_{ij} \equiv e_i e_j$ and $I = e_1 e_2 e_3$ is the **pseudoscalar**. San Jose State University The conventional cyclic ordering $\{e_{12}, e_{23}, e_{31}\}$ is used, noting that $e_{31} = e_3 e_1 = -e_1 e_3$.

A general multivector in G(3) takes the form:

$$M = \alpha + (a_1 e_1 + a_2 e_2 + a_3 e_3) + (b_{12} e_{12} + b_{23} e_{23} + b_{31} e_{31}) + \beta I$$

with eight real coefficients α, a_i, b_{ij}, $\beta \in \mathbb{R}$.

Complete multiplication table for G(3) (row × column):

×	**1**	$\mathbf{e_1}$	$\mathbf{e_2}$	$\mathbf{e_3}$	$\mathbf{e_{12}}$	$\mathbf{e_{23}}$	$\mathbf{e_{31}}$	**I**
1	1	e_1	e_2	e_3	e_{12}	e_{23}	e_{31}	I
$\mathbf{e_1}$	e_1	1	e_{12}	$-e_{31}$	e_2	I	$-e_3$	e_{23}
$\mathbf{e_2}$	e_2	$-e_{12}$	1	e_{23}	$-e_1$	e_3	I	e_{31}
$\mathbf{e_3}$	e_3	e_{31}	$-e_{23}$	1	I	$-e_2$	e_1	e_{12}
$\mathbf{e_{12}}$	e_{12}	$-e_2$	e_1	I	-1	$-e_{31}$	e_{23}	$-e_3$
$\mathbf{e_{23}}$	e_{23}	I	$-e_3$	e_2	e_{31}	-1	$-e_{12}$	$-e_1$
$\mathbf{e_{31}}$	e_{31}	e_3	I	$-e_1$	$-e_{23}$	e_{12}	-1	$-e_2$
I	I	e_{23}	e_{31}	e_{12}	$-e_3$	$-e_1$	$-e_2$	-1

Key derived results from the table:

- **Bivector squares**: $e_{12}^2 = e_1 e_2 e_1 e_2 = -e_1^2 e_2^2 = \mathbf{-1}$. Similarly $e_{23}^2 = e_{31}^2 = \mathbf{-1}$. Bivectors in G(3) behave like imaginary units. arxiv

- **Pseudoscalar square**: $I^2 = (-1)^{(3\cdot 2/2)} \cdot e_1^2 e_2^2 e_3^2 = (-1)^3 \cdot 1 = \mathbf{-1}$.
- **Pseudoscalar commutation**: I **commutes** with all elements of G(3). Instituto de Matemática unicamp For any basis vector e_k, moving e_k through the other two vectors in $I = e_1e_2e_3$ requires exactly two anticommutation swaps, yielding $(-1)^2 = +1$. This property is specific to odd-dimensional spaces.

1.5 The pseudoscalar, duality, and the cross product

The pseudoscalar $I = e_{123}$ provides a **duality map** between grade-k and grade-(3−k) elements via multiplication: Instituto de Matemática unicamp

$$Ie_1 = e_{23}, \qquad Ie_2 = e_{31}, \qquad Ie_3 = e_{12}$$

Since $I^2 = -1$, the inverse is $I^{-1} = -I$, and the inverse duality mapping is:

$$I^{-1}e_{23} = e_1, \qquad I^{-1}e_{31} = e_2, \qquad I^{-1}e_{12} = e_3$$

The traditional **cross product** emerges as the dual of the wedge product: unicamp

$$\boxed{a \times b = -I(a \wedge b) = (a \wedge b)I^{-1}}$$

This reveals the cross product's true nature: it is not a vector operation per se, but the Hodge dual of the naturally bivector-valued wedge product. The cross product exists only in three dimensions precisely because that is where the duality maps bivectors to vectors. arxiv Nicholas Rui

The **even subalgebra** $G^+(3) = \text{span}\{1, e_{12}, e_{23}, e_{31}\}$ is four-dimensional and isomorphic to the **quaternions** $\mathbb{H}$ arXiv Bivector under the identification $e_{12} \leftrightarrow k$, $e_{31} \leftrightarrow j$, $e_{23} \leftrightarrow i$ (with $e_{12}e_{31} = e_{23}$ etc., matching $ijk = -1$ after appropriate sign choices). Uiuc

1.6 Reversion, grade involution, and Clifford conjugation

Three important involutions act on multivectors in geometric algebra.

Reversion (denoted $\tilde{M}$ or $M^\dagger$) reverses the order of all vector factors in every product: Wikipedia Wikipedia

$$(a_1 a_2 \cdots a_k)\tilde{} = a_k \cdots a_2 a_1$$

It is an anti-automorphism: $(AB)\tilde{} = \tilde{B}\tilde{A}$. Wikipedia On a grade-k element, reversion multiplies by the sign factor $(-1)^{(k(k-1)/2)}$: Rdrr.io

Grade	0	1	2	3	4	5
Sign	+	+	−	−	+	+

The pattern repeats with period 4: {+, +, −, −, +, +, −, −, ...}.

Grade involution (denoted $\hat{M}$) negates every vector factor, equivalent to multiplying grade-k parts by $(-1)^k$: Wikipedia Lean Community

$$\hat{M} = \sum_k (-1)^k \langle M \rangle_k$$

It is an automorphism: $(AB)\hat{} = \hat{A}\hat{B}$.

Clifford conjugation (denoted $\bar{M}$) combines both operations: Wikipedia $\bar{M} = (\tilde{M})\hat{} = (\hat{M})\tilde{}$. Wikipedia On grade-k elements, it multiplies by $(-1)^{k(k+1)/2}$, Rdrr.io giving the sign pattern {+, −, −, +, +, −, −, +, ...}.

Summary for G(3):

Component	Grade	$\tilde{M}$	$\hat{M}$	$\bar{M}$
Scalar α	0	+α	+α	+α
Vector a	1	+a	−a	−a
Bivector B	2	−B	+B	−B
Pseudoscalar βI	3	−βI	−βI	+βI

1.7 Rotors, reflections, and the sandwich product

Geometric algebra provides an elegant, dimension-independent framework for reflections and rotations through algebraic products.

Reflection. The reflection of a vector v in the hyperplane perpendicular to unit vector $\hat{n}$ (where $\hat{n}^2 = 1$) is: Uiuc

\boxed{v' = -\hat{n}\,v\,\hat{n}}$

arxiv

This negates the component of v parallel to $\hat{n}$ while preserving perpendicular components. Fiveable To verify: decompose $v = v_\parallel + v_\perp$. Since $v_\parallel \parallel \hat{n}$, they commute: $\hat{n}v_\parallel = v_\parallel\hat{n}$. Since $v_\perp \perp \hat{n}$, they anticommute: $\hat{n}v_\perp = -v_\perp\hat{n}$. Then $-\hat{n}v\hat{n} = -\hat{n}(v_\parallel + v_\perp)\hat{n} = -v_\parallel\hat{n}^2 + v_\perp\hat{n}^2 = -v_\parallel + v_\perp$.

Rotation. A rotation is the composition of two reflections. (arxiv) (Fiveable) If the reflecting hyperplanes have unit normals $\hat{m}$ and $\hat{n}$, the combined transformation is:

$$v' = (\hat{n}\hat{m})\, v\, (\hat{n}\hat{m})\tilde{} = R\, v\, \tilde{R}$$

where **$R = \hat{n}\hat{m}$ is a rotor**. The rotation angle is **twice** the angle between the mirror planes, and the rotation plane is the plane containing $\hat{m}$ and $\hat{n}$. (Bivector)

In general, a rotor for rotation by angle θ in the plane of unit bivector $\hat{B}$ (where $\hat{B}^2 = -1$) is: (unicamp)

$$\boxed{R = e^{-\hat{B}\theta/2} = \cos(\theta/2) - \hat{B}\sin(\theta/2)}$$

The rotation acts via the **sandwich product**:

v' = R\,v\,\tilde{R}$

(Fiveable)

The reverse of the rotor is $\tilde{R} = \cos(\theta/2) + \hat{B}\sin(\theta/2)$, and the normalization condition $R\tilde{R} = 1$ holds automatically. (unicamp) Rotors compose by multiplication: two successive rotations R_1 then R_2 give the combined rotor R_2R_1, and **every multivector** (not just vectors) transforms the same way: $M' = RM\tilde{R}$. (ResearchGate +2)

In G(3), a general rotor has the form $R = a_0 + a_1e_{12} + a_2e_{23} + a_3e_{31}$ (Wikibooks) with the constraint $a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1$, which is precisely the structure of a **unit quaternion**. (Study.com)

Part II: The spacetime algebra G(1,3)

2.1 Basis vectors, metric, and the 16-dimensional structure

The **Spacetime Algebra** (STA) is the geometric algebra $G(1,3) = Cl(1,3)(\mathbb{R})$ of Minkowski spacetime (Wikipedia) with signature (+,−,−,−). It is generated by four orthonormal basis vectors $\{\gamma_0, \gamma_1, \gamma_2, \gamma_3\}$ satisfying:

$$\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\eta_{\mu\nu}, \qquad \eta = \mathrm{diag}(+1, -1, -1, -1)$$

So $\gamma_0^2 = +1$ (timelike), $\gamma_i^2 = -1$ for i = 1,2,3 (spacelike), and $\gamma_\mu\gamma_\nu = -\gamma_\nu\gamma_\mu$ for $\mu \neq \nu$. The algebra is **$2^4 = 16$-dimensional**: (arXiv) (arXiv)

Grade	Basis elements	Count	Name
0	1	1	Scalar
1	$\gamma_0, \gamma_1, \gamma_2, \gamma_3$	4	Vectors
2	$\gamma_{01}, \gamma_{02}, \gamma_{03}, \gamma_{12}, \gamma_{23}, \gamma_{31}$	6	Bivectors
3	$\gamma_{012}, \gamma_{013}, \gamma_{023}, \gamma_{123}$	4	Trivectors
4	$I = \gamma_{0123}$	1	Pseudoscalar

The **STA pseudoscalar** is $I = \gamma_0\gamma_1\gamma_2\gamma_3$ (arXiv) (arXiv) with:

$$I^2 = (-1)^{4\cdot 3/2}\,\gamma_0^2\gamma_1^2\gamma_2^2\gamma_3^2 = (+1)(+1)(-1)(-1)(-1) = -\mathbf{1}$$

Unlike in G(3), the pseudoscalar in STA **anticommutes** with vectors and trivectors (odd-grade elements) and **commutes** with scalars, bivectors, and pseudoscalars (even-grade elements). (Wikipedia) This distinction arises because STA has even dimension n = 4.

The **reciprocal basis** vectors are $\gamma^0 = \gamma_0$ and $\gamma^i = -\gamma_i$ (Wikipedia) (following from the metric: $\gamma^\mu = \eta^{(\mu\mu)}\gamma_\mu$).

2.2 Relative vectors and the spacetime split

The **relative vectors** (or Pauli vectors) are defined as bivectors in STA:

$$\sigma_i = \gamma_i\gamma_0 \qquad (i = 1, 2, 3)$$

Despite being grade-2 objects in STA, the σ_i satisfy the same algebra as the G(3) basis vectors: (arxiv)

$$\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}$$

The σ_i generate the even subalgebra $G^+(1,3) \cong G(3,0)$, establishing a natural embedding of three-dimensional geometry within spacetime. (Wikipedia) The spatial pseudoscalar is $\sigma_1\sigma_2\sigma_3 = I$ (identical to the STA pseudoscalar).

The **spacetime split** of any spacetime vector p relative to observer γ_0 separates temporal and spatial parts: (ResearchGate)

$$p\gamma_0 = p \cdot \gamma_0 + p \wedge \gamma_0 = p^0 + \mathbf{p}$$

where p^0 is a scalar (the temporal component) and $\mathbf{p} = p^i\sigma_i$ is a relative vector (the spatial component).

2.3 Bivector classification: boosts versus rotations

The six STA bivectors split into two geometrically distinct types with different algebraic properties:

- **Timelike bivectors** $\sigma_i = \gamma_i\gamma_0$: these satisfy $\sigma_i^2 = +1$ and generate **Lorentz boosts**. (arxiv)
- **Spacelike bivectors** $\gamma_i\gamma_j = \sigma_i\sigma_j = I\sigma_k$ (for cyclic ijk): these satisfy $(I\sigma_k)^2 = -1$ and generate **spatial rotations**. (arXiv)

This classification is fundamental: the sign of a bivector's square determines whether its exponential produces trigonometric (rotation) or hyperbolic (boost) functions. (Wikipedia)

Part III: Maxwell's equations in geometric algebra

3.1 The four equations in standard vector calculus form

For reference, Maxwell's equations in SI units with standard vector notation are:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss's law)}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad \text{(Gauss's law for magnetism)}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \text{(Faraday's law)}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \qquad \text{(Ampère–Maxwell law)}$$

where $c^2 = 1/(\mu_0\varepsilon_0)$. These four coupled partial differential equations, involving two separate fields E and B with different transformation properties, can be unified into a single equation using geometric algebra.

3.2 The electromagnetic multivector in G(3)

In the Algebra of Physical Space (APS) formulation using G(3), the electric and magnetic fields combine into a single multivector:

$$\boxed{F = \mathbf{E} + Ic\mathbf{B}}$$

where $\mathbf{E} = E_k e_k$ is a **vector** (grade 1) representing the electric field, and $Ic\mathbf{B} = cB_k(Ie_k)$ is a **bivector** (grade 2) representing the magnetic field. Explicitly:

$$F = E_1 e_1 + E_2 e_2 + E_3 e_3 + cB_1 e_{23} + cB_2 e_{31} + cB_3 e_{12}$$

using the duality relations $Ie_1 = e_{23}$, $Ie_2 = e_{31}$, $Ie_3 = e_{12}$. The factor of c ensures consistent dimensions. This construction echoes the classical **Riemann–Silberstein vector** $F = E + icB$ from 1907, but here the imaginary unit i is replaced by the geometric pseudoscalar I, giving it concrete geometric meaning as an oriented volume element. arxiv arXiv

3.3 The single Maxwell equation in G(3)

Define the spatial vector derivative $\nabla = e_k \partial_k$ and the time derivative $\partial_0 = (1/c)\partial/\partial t$. The complete spacetime-aware operator in G(3) is $(\partial_0 + \nabla)$. Maxwell's four equations condense into:

\boxed{\left(\frac{1}{c}\frac{\partial}{\partial t} + \nabla\right)\!F = \frac{\rho}{\varepsilon_0} - \frac{\mathbf{J}}{\varepsilon_0 c}}$

peeterjoot

The right-hand side has a **scalar part** ρ/ε_0 (grade 0) and a **vector part** $-J/(\varepsilon_0 c)$ (grade 1). The left-hand side, when expanded, produces components at all four grades. Setting each grade equal on both sides recovers all four Maxwell equations.

3.4 Grade decomposition recovering all four Maxwell equations

Expanding the left-hand side $(\partial_0 + \nabla)(E + IcB)$ by distributing:

$$\underbrace{\frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}}_{\text{grade 1}} + \underbrace{I\frac{\partial \mathbf{B}}{\partial t}}_{\text{grade 2}} + \underbrace{\nabla \cdot \mathbf{E}}_{\text{grade 0}} + \underbrace{\nabla \wedge \mathbf{E}}_{\text{grade 2}} + \underbrace{\nabla \cdot (Ic\mathbf{B})}_{\text{grade 1}} + \underbrace{\nabla \wedge (Ic\mathbf{B})}_{\text{grade 3}}$$

The key identities for the geometric product of the vector derivative ∇ with a vector and a bivector are:

- $\nabla E = \nabla \cdot E + \nabla \wedge E$, where $\nabla \cdot E$ is the divergence (grade 0) and $\nabla \wedge E = I(\nabla \times E)$ is a bivector (grade 2).
- $\nabla(IcB) = \nabla \cdot (IcB) + \nabla \wedge (IcB)$, where $\nabla \cdot (IcB) = -c(\nabla \times B)$ is a vector (grade 1) and $\nabla \wedge (IcB) = Ic(\nabla \cdot B)$ is a pseudoscalar (grade 3).

Collecting by grade and equating with the right-hand side: Wikipedia

Grade	LHS expression	RHS	Maxwell equation
0 (scalar)	$\nabla\cdot E$	ρ/ε_0	**Gauss's law**
1 (vector)	$(1/c)\partial E/\partial t - c(\nabla\times B)$	$-J/(\varepsilon_0 c)$	**Ampère–Maxwell law**
2 (bivector)	$I(\nabla\times E + \partial B/\partial t)$	0	**Faraday's law**
3 (pseudoscalar)	$Ic(\nabla\cdot B)$	0	**Gauss's law for magnetism**

The grade-1 equation $(1/c)\partial E/\partial t - c(\nabla\times B) = -J/(\varepsilon_0 c)$ becomes, after multiplying by c and using $\mu_0 = 1/(\varepsilon_0 c^2)$: $\nabla\times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t$. The grade-2 equation $I(\nabla\times E + \partial B/\partial t) = 0$ gives $\nabla\times E = -\partial B/\partial t$ since I is invertible.

3.5 The Faraday bivector in STA

In the Spacetime Algebra, the electromagnetic field is a **pure grade-2 bivector**: Aviation Theory Wikipedia

$$F = E^k\sigma_k + cB^k(I\sigma_k)$$

Expanding in the STA basis using $\sigma_k = \gamma_k\gamma_0$ and $I\sigma_k = -\gamma_j\gamma_l$ (for cyclic jkl):

$$F = E_1\gamma_{10} + E_2\gamma_{20} + E_3\gamma_{30} - cB_1\gamma_{23} - cB_2\gamma_{31} - cB_3\gamma_{12}$$

The six independent components of this bivector split naturally: the three **timelike bivectors** γ_{k0} carry the electric field, and the three **spacelike bivectors** γ_{jl} carry the magnetic field. Aviation Theory Under a change of observer (different γ_0'), the same bivector F decomposes into different E′ and B′, reproducing the frame-dependent nature of the electric and magnetic fields. The bivector F itself is **frame-independent**. Chapman University

The relationship to the G(3) formulation uses the spacetime split: right-multiplying by γ_0 gives $F\gamma_0 = E + IcB$, recovering the APS multivector.

3.6 Maxwell's equation in STA and its explicit derivation

The **spacetime vector derivative** is:

$$\nabla = \gamma^\mu\partial_\mu = \gamma_0\frac{\partial}{\partial(ct)} - \gamma_1\frac{\partial}{\partial x^1} - \gamma_2\frac{\partial}{\partial x^2} - \gamma_3\frac{\partial}{\partial x^3}$$

where $\gamma^0 = \gamma_0$ and $\gamma^i = -\gamma_i$. The **4-current density** is the spacetime vector:

$$J = c\rho\,\gamma_0 + J^1\gamma_1 + J^2\gamma_2 + J^3\gamma_3$$

All four Maxwell equations are encoded in the single STA equation:

$$\boxed{\nabla F = \frac{J}{\varepsilon_0 c}}$$

Since ∇ is grade-1 and F is grade-2, their geometric product produces:

$$\nabla F = \underbrace{\nabla \cdot F}_{\text{grade 1 (vector)}} + \underbrace{\nabla \wedge F}_{\text{grade 3 (trivector)}}$$

The **sourced Maxwell equations** come from the vector part: $\nabla\cdot F = J/(\varepsilon_0 c)$. This single vector equation encodes both Gauss's law (the γ_0 component gives $\nabla\cdot E = \rho/\varepsilon_0$) and the Ampère–Maxwell law (the γ_k components give $\nabla\times B = \mu_0 J + \mu_0\varepsilon_0\,\partial E/\partial t$).

The **source-free Maxwell equations** come from the trivector part: $\nabla\wedge F = 0$. This encodes both Gauss's law for magnetism (the γ_{123} component gives $\nabla\cdot B = 0$) and Faraday's law (the γ_{0jk} components give $\nabla\times E = -\partial B/\partial t$).

In vacuum (J = 0), the equation simplifies to $\nabla\mathbf{F} = \mathbf{0}$, and the separation $\nabla\cdot F = 0$ and $\nabla\wedge F = 0$ shows the source-free field satisfies both sets of equations simultaneously.

3.7 Electromagnetic duality

The **duality rotation** F → Fe^(Iθ) for real parameter θ continuously transforms between electric and magnetic content. For the discrete duality θ = π/2:

$$F \to FI = (E + IcB)I = EI + I^2 cB = -cB + IE$$

This exchanges E ↔ −cB and cB ↔ E, the classical electromagnetic duality. The vacuum Maxwell equation ∇F = 0 is **invariant** under this rotation since I commutes with the bivector ∇F:

$$\nabla(Fe^{I\theta}) = (\nabla F)e^{I\theta} = 0$$

This duality symmetry is broken by the presence of electric sources. It would be restored only if magnetic monopoles existed, replacing J with a complex source j = j_e + Ij_m.

The **Lorentz invariants** of the field emerge from the square of F:

$$F^2 = (|\mathbf{E}|^2 - c^2|\mathbf{B}|^2) + 2cI(\mathbf{E}\cdot\mathbf{B})$$

The scalar part $|E|^2 - c^2|B|^2$ and the pseudoscalar part $2c(E\cdot B)$ are both Lorentz invariant.

Part IV: Potentials, gauge theory, and the wave equation

4.1 The electromagnetic 4-potential and $F = \nabla\wedge A$

The electromagnetic 4-potential is a spacetime vector (grade-1 element in STA):

$$A = \frac{\phi}{c}\gamma_0 + A^1\gamma_1 + A^2\gamma_2 + A^3\gamma_3$$

where φ is the scalar potential and A^i are the magnetic vector potential components. The field bivector is obtained as the **exterior derivative** of the potential:

$$\boxed{F = \nabla \wedge A}$$

This definition automatically satisfies the source-free Maxwell equations, since the outer product of the derivative operator with itself vanishes by the symmetry of partial derivatives:

$$\nabla \wedge F = \nabla \wedge (\nabla \wedge A) = 0$$

This identity encodes $\nabla\cdot B = 0$ and $\nabla\times E = -\partial B/\partial t$ with no further conditions required.

4.2 Gauge freedom and the Lorenz condition

The geometric product of ∇ with A decomposes as:

$$\nabla A = \nabla \cdot A + \nabla \wedge A = (\nabla \cdot A) + F$$

The scalar quantity $\nabla\cdot A$ is **gauge-dependent**: the transformation $A \to A + \nabla\chi$ for any scalar function χ leaves $F = \nabla\wedge A$ unchanged (since $\nabla\wedge\nabla\chi = 0$) but shifts $\nabla\cdot A \to \nabla\cdot A + \nabla^2\chi$. The standard gauge choices are:

$$\text{Lorenz gauge:} \quad \nabla \cdot A = 0 \qquad (= \partial_\mu A^\mu = 0)$$

$$\text{Coulomb gauge:} \quad \nabla_{\text{spatial}} \cdot \mathbf{A} = 0$$

In the **Lorenz gauge**, the geometric product simplifies beautifully to $F = \nabla A$ (the full geometric product, not just the wedge part).

4.3 The wave equation

Substituting $F = \nabla \wedge A = \nabla A - (\nabla \cdot A)$ into Maxwell's equation $\nabla F = J/(\varepsilon_0 c)$:

$$\nabla(\nabla A) - \nabla(\nabla \cdot A) = \frac{J}{\varepsilon_0 c}$$

Since $\nabla(\nabla A) = \nabla^2 A$ (the d'Alembertian), imposing the Lorenz gauge $\nabla \cdot A = 0$ gives:

$$\boxed{\nabla^2 A = \Box A = \frac{J}{\varepsilon_0 c}}$$

where $\nabla^2 = \nabla \cdot \nabla = \Box = \partial^2/(c^2\partial t^2) - \partial^2/\partial x^2 - \partial^2/\partial y^2 - \partial^2/\partial z^2$ is the **d'Alembertian** (wave operator). This is the covariant wave equation for the 4-potential, equivalent to the pair of scalar equations $\Box\varphi = \rho/\varepsilon_0$ and $\Box A = \mu_0 J$ in component form.

Part V: Advanced topics

5.1 The electromagnetic stress-energy tensor

In STA, the stress-energy tensor is expressed as a **vector-valued linear function** of a vector argument n, rather than as a rank-2 tensor with indices:

$$\boxed{T(n) = -\frac{\varepsilon_0}{2} F n F = \frac{\varepsilon_0}{2} F n \tilde{F}}$$

The second equality uses the fact that for any bivector F, the reversion $\tilde{F} = -F$ (since reversion negates grade-2 elements). The function T(n) maps a spacetime vector n to a spacetime vector T(n), encoding energy density, momentum density, and stress.

Evaluating for specific arguments in the γ_0 frame:

- **Energy density** (from T^{00}): $u = \langle T(\gamma_0)\gamma_0\rangle_0 = (\varepsilon_0/2)(E^2 + c^2B^2) = (1/2)(\varepsilon_0 E^2 + B^2/\mu_0)$
- **Poynting vector** (from $T^0{}_i$): S/c extracts from the σ_i components of $T(\gamma_0)$, giving $\mathbf{S} = (1/\mu_0)E\times B$
- **Maxwell stress tensor** (from T^{ij}): $\sigma_{ij} = \varepsilon_0 E_i E_j + (1/\mu_0)B_i B_j - ½(\varepsilon_0 E^2 + B^2/\mu_0)\delta_{ij}$

The stress-energy tensor is **traceless** ($T^{\mu}{}_{\mu} = 0$), reflecting the conformal symmetry of classical electromagnetism. Conservation of energy-momentum is expressed as $\partial_\mu T^{\mu\nu} = F^{\nu\lambda} J_\lambda$, which in STA becomes $\nabla \cdot T(n) = (F \cdot J) \cdot n$ for the force density exerted on charges.

5.2 Plane wave solutions in GA notation

For a **linearly polarized monochromatic plane wave** propagating along e_3 with E polarized along e_1:

$$\mathbf{E} = E_0 \cos(kz - \omega t)\, e_1, \qquad \mathbf{B} = \frac{E_0}{c} \cos(kz - \omega t)\, e_2$$

The GA field multivector is:

$$F = E_0 \cos(kz - \omega t)\, (e_1 - e_{13})$$

where $e_{13} = e_1 e_3$ is a bivector. The factor $(e_1 - e_{13}) = e_1(1 - e_3)$ encodes both the electric polarization direction and the magnetic field plane. This can be written more compactly as $F = E_0 \cos(kz - \omega t) \cdot e_1(1 - e_3)$.

In STA, the plane wave uses the **null wave vector** $k = (\omega/c)\gamma_0 + k^i\gamma_i$ satisfying $k^2 = 0$ (giving the dispersion relation $\omega = c|k|$). The field bivector takes the form:

$$F = f_0\, e^{I\Phi}, \qquad \Phi = k \cdot x = \omega t - \mathbf{k} \cdot \mathbf{r}$$

where f_0 is a constant bivector amplitude. The factor $e^{I\Phi}$ performs a **duality rotation** that oscillates between electric and magnetic content.

Polarization states map naturally onto duality rotations. Linear polarization corresponds to a fixed bivector direction modulated by $\cos(\Phi)$. Circular polarization arises from a $\pi/4$ duality rotation: $F = f_0 \exp(\pm I\pi/4) \exp(I\Phi)$, producing a 90° phase shift between E and B that generates helicity.

5.3 Lorentz transformations as rotors in STA

Any proper orthochronous Lorentz transformation of a spacetime vector a is:

$$a' = R\, a\, \tilde{R}$$

where R is an even-grade multivector (rotor) satisfying $R\tilde{R} = 1$. The six-parameter Lorentz group is generated by six independent bivectors.

Spatial rotation by angle θ about axis $\hat{n}$ uses a spacelike bivector (which squares to −1):

$$R_{\text{rot}} = e^{-I\sigma_{\hat{n}}\,\theta/2} = \cos(\theta/2) - I\sigma_{\hat{n}} \sin(\theta/2)$$

where $I\sigma_{\hat{n}}$ is the bivector dual to $\hat{n}$ in the spatial subalgebra. For rotation about σ_3: $R = \exp(-I\sigma_3\,\theta/2) = \cos(\theta/2) - I\sigma_3 \sin(\theta/2)$, using the fact that $(I\sigma_3)^2 = -1$.

Lorentz boost with rapidity α in direction $\hat{n}$ uses a timelike bivector (which squares to +1):

$$R_{\text{boost}} = e^{-\sigma_{\hat{n}}\,\alpha/2} = \cosh(\alpha/2) - \sigma_{\hat{n}} \sinh(\alpha/2)$$

where $\tanh \alpha = v/c$. The hyperbolic functions appear because $\sigma_{\hat{n}}^2 = +1$.

The **electromagnetic field transforms** under a Lorentz boost L as $F' = LF\tilde{L}$. Expanding for a boost along σ_1 with rapidity α yields the standard transformation rules:

$$E'_{\parallel} = E_{\parallel}, \qquad B'_{\parallel} = B_{\parallel}$$

$$E'_{\perp} = \gamma(E_{\perp} + \mathbf{v} \times \mathbf{B}), \qquad B'_{\perp} = \gamma\left(B_{\perp} - \frac{\mathbf{v}}{c^2} \times \mathbf{E}\right)$$

where $\gamma = \cosh \alpha$. The GA derivation requires only expanding the sandwich product and applying the σ_i commutation rules, far more economical than the index-laden tensor approach.

5.4 Spinors as even multivectors

In STA, a **Dirac spinor** is represented as an even-grade multivector (scalar + bivector + pseudoscalar), with the canonical Hestenes decomposition:

$$\psi = \rho^{1/2}\, e^{I\beta/2}\, R$$

where **ρ** is a positive scalar related to the probability density, **β** is the Yvon-Takabayasi angle ($0 \le \beta \le \pi$), and **R** is a rotor ($R\tilde{R} = 1$) encoding the local Lorentz frame. This decomposition gives **8 real degrees of freedom**, matching the 4 complex components of a standard Dirac column spinor.

The **Dirac equation** in STA (the Hestenes form) is:

$$\nabla\psi\, I\sigma_3 - eA\psi = m\psi\gamma_0$$

where the factor $I\sigma_3 = \gamma_2\gamma_1$ plays the role of the imaginary unit i of quantum mechanics, now given geometric meaning as the **spin plane** bivector. The physical observables extract directly from ψ:

- **Probability current**: $J = \psi\gamma_0\tilde{\psi}$ (a conserved timelike vector)
- **Spin bivector**: $S = (\hbar/2)\psi\sigma_3\tilde{\psi}$

The correspondence between column spinors and even multivectors maps the abstract Hilbert space formalism onto concrete geometric objects with direct physical interpretation.

5.5 How GA compares to differential forms

Both geometric algebra and the calculus of differential forms provide coordinate-free frameworks for electromagnetism, but they differ in fundamental ways.

In **differential forms**, the electromagnetic field is a 2-form $F = \frac{1}{2}F_{\mu\nu}\, dx^\mu \wedge dx^\nu$, and Maxwell's equations require **two separate equations**:

$$dF = 0 \quad \text{(Bianchi identity)}, \qquad d{\star}F = {\star}J \quad \text{(dynamical equation)}$$

where d is the exterior derivative and $\star$ is the Hodge star operator. In **geometric algebra**, these unify into the single equation $\nabla F = J/(\varepsilon_0 c)$, because the geometric product automatically produces both the inner product (which yields the sourced equations) and the outer product (which yields the source-free equations).

The core structural difference is the **geometric product** itself. Differential forms possess only the wedge product, which is not invertible. The geometric product $ab = a\cdot b + a\wedge b$ is invertible for nonzero elements, enabling division, Green's function methods, and direct algebraic manipulation that forms cannot support. The **Hodge star** $\star$ in forms, which requires a metric and orientation, corresponds to simple multiplication by the pseudoscalar I in GA — a purely algebraic operation.

For **rotations and Lorentz transformations**, GA provides built-in rotors ($R = e^{(-B\theta/2)}$) with the sandwich product $v' = Rv\tilde{R}$, requiring no external Lie group machinery. Forms need separate constructions for this purpose. For **spinors**, GA accommodates them naturally as even-grade multivectors, whereas differential forms require additional bundle structures.

Differential forms hold advantages in **pure mathematics**: the exterior derivative needs no metric (only differentiable structure), making forms natural on general manifolds. Stokes' theorem $\int_{\partial M} \omega = \int_M d\omega$ and de Rham cohomology connect directly to topology. Forms also enjoy wider adoption in general relativity and gauge theory literature.

The comparison reduces to a tradeoff: GA provides a richer algebraic structure that unifies more physics into fewer equations, while differential forms provide a more minimal framework with deeper connections to topology and manifold theory.

Part VI: Worked examples with full derivations

6.1 Deriving Gauss's law from the single GA Maxwell equation

Goal: Extract $\nabla\cdot E = \rho/\varepsilon_0$ from $(1/c\, \partial_t + \nabla)(E + IcB) = \rho/\varepsilon_0 - J/(\varepsilon_0 c)$.

Step 1. Expand the left-hand side by distribution:

$$(1/c\,\partial_t + \nabla)(E + IcB) = \frac{1}{c}\partial_t E + I\partial_t B + \nabla E + \nabla(IcB)$$

Step 2. Expand ∇E using the geometric product of two vectors:

$$\nabla E = (e_k\partial_k)(E_j e_j) = \sum_{k,j}(\partial_k E_j)\, e_k e_j$$

Separate diagonal (k = j) and off-diagonal (k ≠ j) terms using $e_k e_j = \delta_{kj} + e_k \wedge e_j$:

$$\nabla E = \sum_k (\partial_k E_k)\underbrace{e_k e_k}_{=1} + \sum_{k\neq j}(\partial_k E_j)\underbrace{e_k e_j}_{=e_{kj}}$$

$$= \underbrace{(\partial_1 E_1 + \partial_2 E_2 + \partial_3 E_3)}_{\text{grade 0: } \nabla\cdot E} + \underbrace{\sum_{k\neq j}(\partial_k E_j)\, e_{kj}}_{\text{grade 2: } \nabla\wedge E}$$

Step 3. The grade-0 part of the entire left-hand side comes only from ∇·E (the terms (1/c)∂_tE, I∂_tB, and ∇(IcB) contribute no scalars). Therefore the grade-0 equation is:

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

This is Gauss's law. ■

6.2 Deriving Faraday's law from the single GA Maxwell equation

Goal: Extract ∇×E = −∂B/∂t from the grade-2 part.

Step 1. Identify all grade-2 contributions from the left-hand side:

- From (1/c)∂_tE: this is a vector (grade 1), contributes nothing to grade 2.
- From I∂_tB: since B is a vector and I maps vectors to bivectors, this is **I(∂B/∂t)**, a bivector (grade 2).
- From ∇E: the grade-2 part is ∇∧E = I(∇×E), a bivector.
- From ∇(IcB): expanding the product of a vector with a bivector, the grade-1 part is ∇·(IcB) and the grade-3 part is ∇∧(IcB). Neither contributes to grade 2.

Step 2. Collect the grade-2 terms:

$$\langle \text{LHS} \rangle_2 = I(\nabla \times E) + I\frac{\partial B}{\partial t} = I\left(\nabla \times E + \frac{\partial B}{\partial t}\right)$$

Step 3. The right-hand side $\rho/\varepsilon_0 - J/(\varepsilon_0 c)$ contains only grades 0 and 1, so the grade-2 equation is:

$$I\left(\nabla \times E + \frac{\partial B}{\partial t}\right) = 0$$

Step 4. Since I is invertible ($I^{-1} = -I$ because $I^2 = -1$), multiply both sides by I^{-1}:

$$\nabla \times E + \frac{\partial B}{\partial t} = 0 \quad \Longrightarrow \quad \nabla \times E = -\frac{\partial B}{\partial t}$$

This is Faraday's law. ■

Note: The remaining two equations follow identically. The grade-1 equation gives $(1/c)\partial E/\partial t - c(\nabla \times B) = -J/(\varepsilon_0 c)$, which simplifies to the Ampère–Maxwell law. The grade-3 equation gives $Ic(\nabla \cdot B) = 0$, which (since $Ic \neq 0$) yields $\nabla \cdot B = 0$.

6.3 Expressing a plane electromagnetic wave in GA and verifying Maxwell's equation

Setup: A linearly polarized plane wave propagating along e_3 with E along e_1:

$$E = E_0 \cos(kz - \omega t)\, e_1, \qquad B = \frac{E_0}{c} \cos(kz - \omega t)\, e_2$$

Step 1. Construct the field multivector:

$$cB = E_0 \cos(kz - \omega t)\, e_2$$

$$IcB = E_0 \cos(kz - \omega t)\, Ie_2 = E_0 \cos(kz - \omega t)\, e_{31}$$

where $Ie_2 = e_{123} \cdot e_2 = e_{31}$ (verified: $e_1e_2e_3e_2 = -e_1e_3 = e_3e_1 = e_{31}$).

Therefore:

$$F = E_0 \cos(kz - \omega t)\, (e_1 + e_{31})$$

Note that $e_{31} = e_3e_1 = -e_1e_3 = -e_{13}$, so we can also write $F = E_0 \cos(kz - \omega t)(e_1 - e_{13})$.

Step 2. Verify the source-free equation $(1/c\ \partial_t + \nabla)F = 0$:

Let $\xi = kz - \omega t$ and $f(\xi) = E_0 \cos \xi$, so $f'(\xi) = -E_0 \sin \xi$.

Time derivative: $(1/c)\partial_t F = (1/c)(-\omega)f'(\xi)$ is incorrect — let me redo. Actually $f(\xi) = E_0\cos(\xi)$, so $\partial f/\partial t = -E_0\sin(\xi)\cdot(-\omega) = \omega E_0\sin(\xi)$. Wait — $\partial\xi/\partial t = -\omega$, so $\partial f/\partial t = f'(\xi)\cdot(-\omega)$. And $f'(\xi) = -E_0\sin\xi$. So $\partial f/\partial t = E_0\omega \sin\xi$.

Let me use a cleaner approach. Write $F = f(\xi)(e_1 + e_{31})$ where $\xi = kz - \omega t$.

$$(1/c)\,\partial_t F = \frac{1}{c}\frac{\partial f}{\partial t}(e_1 + e_{31}) = \frac{-\omega}{c} f'(\xi)(e_1 + e_{31}) \cdot \frac{1}{f'} = \frac{1}{c}(-\omega) f'(\xi)(e_1 + e_{31})$$

Wait, let me just be explicit. We have $\partial F/\partial t = (df/d\xi)(\partial\xi/\partial t)(e_1 + e_{31}) = f'(\xi)(-\omega)(e_1 + e_{31})$.

So $\mathbf{(1/c)\partial_tF = -(\omega/c)f'(\xi)(e_1 + e_{31}) = -kf'(\xi)(e_1 + e_{31})}$ using $\omega = ck$.

Spatial derivative: Since F depends only on z, $\nabla F = e_3\partial_3 F = e_3 \cdot kf'(\xi)(e_1 + e_{31})$.

Now compute the products:

- $e_3e_1 = -e_1e_3 = -e_{13} = e_{31}$ (a bivector)
- $e_3e_{31} = e_3(e_3e_1) = (e_3^2)e_1 = e_1$ (a vector)

So $e_3(e_1 + e_{31}) = e_{31} + e_1 = (e_1 + e_{31})$.

Therefore: $\mathbf{\nabla F = kf'(\xi)(e_1 + e_{31})}$.

Step 3. Sum:

$$(1/c)\,\partial_t F + \nabla F = -kf'(\xi)(e_1 + e_{31}) + kf'(\xi)(e_1 + e_{31}) = \mathbf{0} \quad \checkmark$$

The cancellation occurs because e_3 acting on $(e_1 + e_{31})$ returns $(e_1 + e_{31})$ itself — an eigenvalue-1 property reflecting the self-reinforcing nature of electromagnetic wave propagation. ■

6.4 Rotation of e_1 by angle θ around the e_3 axis using a rotor

Step 1. Construct the rotor. The rotation plane is e_1e_2, so the unit bivector is $\hat{B} = e_{12}$ with $e_{12}^2 = -1$:

$$R = e^{-e_{12}\theta/2} = \cos(\theta/2) - e_{12}\sin(\theta/2)$$

Write $c \equiv \cos(\theta/2)$ and $s \equiv \sin(\theta/2)$ for brevity.

Step 2. Construct the reverse: $\tilde{R} = \cos(\theta/2) + e_{12}\sin(\theta/2) = c + e_{12}\, s$.

Verify normalization: $R\tilde{R} = (c - e_{12}s)(c + e_{12}s) = c^2 - e_{12}^2 s^2 = c^2 + s^2 = 1$. ✓

Step 3. Compute Re_1:

$$Re_1 = (c - e_{12}s)e_1 = ce_1 - se_{12}e_1$$

Evaluate $e_{12}e_1 = (e_1e_2)e_1 = e_1(e_2e_1) = e_1(-e_1e_2) = -e_1^2e_2 = \mathbf{-e_2}$.

$$Re_1 = ce_1 + se_2$$

Step 4. Compute $(Re_1)\tilde{R}$:

$$(ce_1 + se_2)(c + e_{12}s)$$

Expand all four terms:

- $ce_1 \cdot c = c^2e_1$
- $ce_1 \cdot e_{12}s = cs \cdot e_1e_{12} = cs \cdot (e_1e_1)e_2 = cs \cdot e_2 = \mathbf{cse_2}$
- $se_2 \cdot c = cse_2 = \mathbf{cse_2}$
- $se_2 \cdot e_{12}s = s^2 \cdot e_2e_{12} = s^2 \cdot (e_2e_1)e_2 = s^2 \cdot (-e_1)e_2^2 = \mathbf{-s^2e_1}$

Step 5. Collect terms:

$$v' = (c^2 - s^2)e_1 + 2cs\, e_2$$

Step 6. Apply double-angle identities:

$$v' = \cos\theta\, e_1 + \sin\theta\, e_2$$

This is the vector e_1 rotated counterclockwise by angle θ in the e_1e_2 plane. ■

Verification: At $\theta = \pi/2$, $v' = e_2$ ✓. At $\theta = \pi$, $v' = -e_1$ ✓. At $\theta = 0$, $v' = e_1$ ✓.

6.5 Reflection of a vector in a plane

Setup: Reflect $v = e_1 + 2e_2$ in the plane perpendicular to $n = e_1$ (the yz-plane).

Formula: $v' = -nvn$ for unit n (negates the component along n, preserves perpendicular components).

Step 1. Compute $vn = (e_1 + 2e_2)e_1$:

$$vn = e_1e_1 + 2e_2e_1 = 1 + 2(-e_{12}) = 1 - 2e_{12}$$

Step 2. Compute n(vn) = $e_1(1 - 2e_{12})$:

$$e_1(1 - 2e_{12}) = e_1 - 2e_1e_{12} = e_1 - 2(e_1e_1)e_2 = e_1 - 2e_2$$

Step 3. Apply the minus sign:

$$v' = -(e_1 - 2e_2) = -e_1 + 2e_2$$

The e_1 component flipped sign ($1 \to -1$) and the e_2 component is preserved ($2 \to 2$), exactly as expected for reflection through the yz-plane. ■

Conclusion: why geometric algebra matters for electromagnetism

The reformulation of Maxwell's equations in geometric algebra is not merely notational convenience — it reveals structural insights invisible in the vector calculus formulation. The unification of E and B into a single bivector F exposes electromagnetic duality as multiplication by the pseudoscalar I, frames Lorentz invariants as the scalar and pseudoscalar parts of F^2, and reduces all four Maxwell equations to the single algebraic statement $\nabla F = J/(\varepsilon_0 c)$. The grade decomposition of this equation provides a systematic, mechanical procedure for recovering individual field equations, replacing ad hoc vector identities with a uniform algebraic framework.

The rotor formalism eliminates the artificial distinction between rotations and boosts: both are exponentials of bivectors, differing only in whether the bivector squares to −1 or +1. The sandwich product $v' = Rv\tilde{R}$ applies uniformly to vectors, bivectors, and spinors, replacing the zoo of transformation rules (rotation matrices, Lorentz transformation matrices, spinor representations) with a single operation. The natural accommodation of spinors as even-grade multivectors, with the Dirac equation taking the form $\nabla\psi I\sigma_3 - eA\psi = m\psi\gamma_0$, gives geometric meaning to the imaginary unit of quantum mechanics as the spin-plane bivector.

For practical computation, GA offers the invertibility of the geometric product (enabling algebraic division by vectors), the direct construction of Green's functions for the first-order operator ∇, and the ability to write gauge theory compactly as $F = \nabla \wedge A$ with the wave equation $\Box A = J$. The cost is learning a new algebraic language, but the payoff is a unified, coordinate-free framework in which classical electromagnetism, special relativity, and quantum spinor theory all speak the same mathematical language.